Source: White.Rainforest ™ · 易雨白林. on Unsplash

MANY HUMANS IN MANY LOOPS

CARLY RICHMOND

AGENDA

- What is HITL?
- Reinforcement Learning with Feedback
- Context Engineering
- Alert Fatigue management

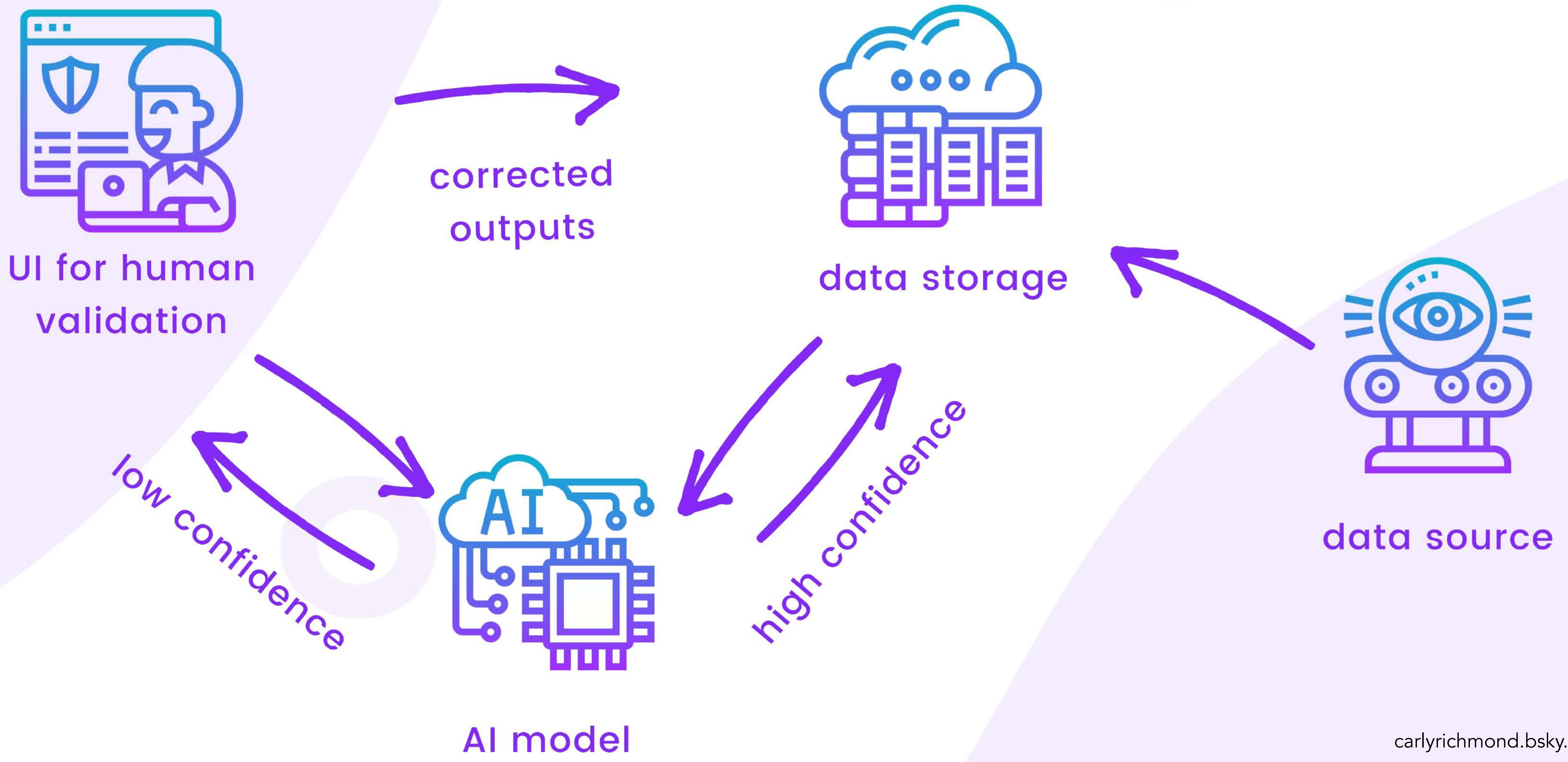
ABOUT ME

- Developer Advocate Lead @ elastic
- Frontend Engineer, Speaker & Blogger

SCAN ME

carlyrichmond.bsky.social

how it works



UI for human validation
corrected outputs
data storage
data source
low confidence
high confidence
AI
AI model

Search logs
Build and push recomme image
74 #17 extracting sha256:
5 #17 extracting sha256
#17 extracting sha25
#17 extracting sh
aef8b136106b41f5cd30530992ed6ad5f3b48b1ce9516084d69ba3cea3 done
0d88c5aa2af662a89e63ecc1ae5e7b2f21bcca0f4cbfb8dbec86f561b8
80d88c5aa2af662a89e63ecc1ae5e7b2f21bcca0f4cbfb8dbec86f561b8 0.4s done
6ec31be71857d4950bb7295b620708d18f5ca3d726106961e770a01d49de
766ec31be71857d4950bb7295b620708d18f5ca3d726106961e770a01d49de done
:61dc7907f21620c0082381c405790ff0e559545427f9092294cb3266b3222456 done
280 sha256:dc464a433611073dd2fcaf168e8b7f8dcb97346210c70da0ac8ced592de1b515 done
281 #17 extracting sha256:78f5ebbb9b61490dbf4813702fed1cf945a08f0e0fb5518e716bf1c2ac8e987d done
282 #17 extracting sha256:83c2237badfc793d0dd44551e3f6027b3b94162f98a448287f12ed3ec0ed49f0 done
283 #17 DONE 2.3s
284 #18 [stage-1 8/8] COPY ./src/recommendation/recommendation_server.py recommendation_server.py
285 #18 DONE 0.1s
286 #19 exporting to image
287 #19 exporting layers 0.0s done
288 #19 exporting manifest sha256:f811b91734f4a8d3ee9b2f94c550e2bef050b424cf2102e839c9d688cfb09439 done
289 #19 exporting config sha256:34f2537cc04598ad43c0a9637be8adb43a7cb3112e8bebeae37c93d76d890870 done
290 #19 exporting attestation manifest sha256:fbe9d07447b353045e69376b134b2a3e995660633045c7439b5500bd8b026302 don
291 #19 exporting manifest list sha256:9ab3fa2b9e71df607a4d3a6724a7c56685bfd59f3381bb5ae45d45eaf6f9e456 done
292 #19 pushing layers
295 #19 ...
296 #20 [auth] alexanderwert/demo:pull,push token for ghcr.io
297 #20 DONE 0.0s
300 #19 exporting to image

IT WORKS!

Cluster: gke_elastic-observability_europe-west3_alex-w-demo-otel-kubecon
User: gke_elastic-observability_europe-west3_alex-w-demo-otel-kubecon
K9s Rev: v0.40.8 ⚡v0.50.18
K8s Rev: v1.34.3-gke.1444000
CPU: 41%
MEM: 61%

<1> default

<ctrl-d>	Delete	<l>	Logs	<f>	Show PortForward
<d>	Describe	<p>	Logs Previous	<t>	Transfer
<e>	Edit	<shift-f>	Port-Forward	<y>	YAML
<?>	Help	<z>	Sanitize		
<shift-j>	Jump Owner	<s>	Shell		

v1/pods(default)[25]

NAME↑	PF	READY	STATUS	RESTARTS	CPU	MEM	%CPU/R	%CPU/L	%MEM/R	%MEM/L	IP	NODE	AGE
accounting-bcccb46f-4b9lv	●	1/1	Running	0	4	147	n/a	n/a	98	98	10.48.3.215	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-qt85	147m
ad-5d9b7fcf7b-69d5b	●	1/1	Running	0	3	238	n/a	n/a	79	79	10.48.3.216	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-qt85	147m
cart-787df469f6-xh2rl	●	1/1	Running	0	4	74	n/a	n/a	46	46	10.48.3.217	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-qt85	147m
checkout-767b744d46-g7rlj	●	1/1	Running	1	2	15	n/a	n/a	77	77	10.48.3.218	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-qt85	147m
currency-f7fb8c49f-dcj2j	●	1/1	Running	0	2	7	n/a	n/a	39	39	10.48.3.219	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-qt85	147m
email-d6bf57b4d-62cqz	●	1/1	Running	0	2	66	n/a	n/a	66	66	10.48.3.220	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-qt85	147m
flagd-58ddc57565-rmsk9	●	2/2	Running	0	3	180	n/a	n/a	55	55	10.48.2.39	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	147m
fraud-detection-69fb6f45bb-jqzc6	●	1/1	Running	0	7	234	n/a	n/a	78	78	10.48.2.40	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	147m
frontend-dcb9588d9-spvxm	●	1/1	Running	0	25	120	n/a	n/a	48	48	10.48.3.221	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-qt85	147m
frontend-proxy-6fd5fb8d6d-62pcc	Ⓕ	1/1	Running	0	9	20	n/a	n/a	32	32	10.48.2.41	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	147m
image-provider-56b5b557cc-z7jpz	●	1/1	Running	0	1	12	n/a	n/a	24	24	10.48.0.187	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-2sfm	147m
kafka-8bcd9cdb8-mmvqc	●	1/1	Running	0	17	544	n/a	n/a	90	90	10.48.0.188	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-2sfm	147m
llm-c558f6b46-8n4t2	●	1/1	Running	0	21	70	n/a	n/a	n/a	n/a	10.48.3.222	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-qt85	147m
load-generator-69d5766587-z9xtb	●	1/1	Running	0	1549	1178	n/a	n/a	78	78	10.48.2.46	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	147m
otel-collector-agent-858j9	●	1/1	Running	0	34	103	n/a	n/a	51	51	10.48.3.183	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-qt85	23h
otel-collector-agent-fxk8k	●	1/1	Running	0	40	112	n/a	n/a	56	56	10.48.0.149	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-2sfm	23h
otel-collector-agent-pzsds	●	1/1	Running	0	74	110	n/a	n/a	55	55	10.48.2.244	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	23h
payment-57db774765-2ppjp	●	1/1	Running	0	23	96	n/a	n/a	69	69	10.48.2.42	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	147m
postgresql-6c9866bdf-9zg5f	●	1/1	Running	0	11	70	n/a	n/a	70	70	10.48.2.7	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	6d17h
product-catalog-84c5779894-ffpnv	●	1/1	Running	0	7	12	n/a	n/a	61	61	10.48.2.43	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	147m
product-reviews-7d6f59f467-49xs9	●	1/1	Running	0	23	73	n/a	n/a	73	73	10.48.2.44	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	147m
quote-bf89f65c4-vq286	●	1/1	Running	0	1	23	n/a	n/a	58	58	10.48.0.189	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-2sfm	147m
recommendation-9bbf4bf6f-jcmsd	●	1/1	Running	0	15	52	n/a	n/a	10	10	10.48.0.190	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-2sfm	147m
shipping-584c65cfb5-vd26b	●	1/1	Running	0	1	6	n/a	n/a	34	34	10.48.2.45	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	147m
valkey-cart-7784949c-jk2g5	●	1/1	Running	0	4	3	n/a	n/a	16	16	10.48.2.14	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	6d17h

D
k8-otel-demo
Dashboards
[Kubernetes OTel] Cluster Overview
Managed
AI Agent
CR
Observability
Discover
Dashboards
Workflows
Alerts
Cases
SLOs
Streams
Applications
More
elastic
Share
Export
Filter your data using KQL syntax
Last 15 minutes
Refresh
Cluster Any
Node Any
Namespace Any
Cluster overview
Nodes
1
Namespaces
1
CPU Usage
Cores
10.0
8.0
6.0
4.0
2.0
0.0
17:05
May 27, 2026
17:10
17:15
Allocatable
Requests
Limits
Memory Usage
Bytes
8.00GB
6.00GB
4.00GB
2.00GB
0.00B
Node overview
Nodes Status
Ready 1
Top CPU intensive Nodes
% CPU
14.00%
12.00%
10.00%
8.00%
6.00%
4.00%
2.00%
0.00%
Top Memory intensive Nodes
% Memory
50.00%
40.00%
30.00%
20.00%
10.00%
0.00%
Top Filesystem Utilization intensive Nodes
% FS usage
10.00%
8.00%
6.00%
4.00%
2.00%
0.00%
Workload overview
Container Restarts by Pod and Reason
Pods
Running 1
Deployments
Ready 23
DaemonSets
Ready 2
StatefulSets
No results found
ReplicaSets
Ready 23
Top CPU intensive pods per Node
Top CPU intensive pods per Pod Limit
Top Memory intensive pods per Node
Top Memory intensive pods per Pod Limit

...MAYBE NOT!

OTel Pal APP 2:34 PM

" 🚨 **Latency Alert: frontend-proxy GET /api/recommendations**

Problem: The `frontend-proxy` service is experiencing elevated latency (318 ms avg, threshold: 200 ms) on the `GET /api/recommendations` transaction.

Root Cause: Commit `a5343cd` in the `recommendation` service introduced a faulty caching function that causes exponential list growth and always executes the expensive path on every request, resulting in unbounded memory consumption and CPU overhead.

Case: https://otel-demo-d90c84.kb.europe-west3.gcp.elastic.cloud/app/observability/cases/6a5f737d-ba66-4f4d-bfb3-59ac3d37d182

Automatic Remediation: A PR has been created to fix this issue:
https://github.com/AlexanderWert/opentelemetry-demo/pull/7

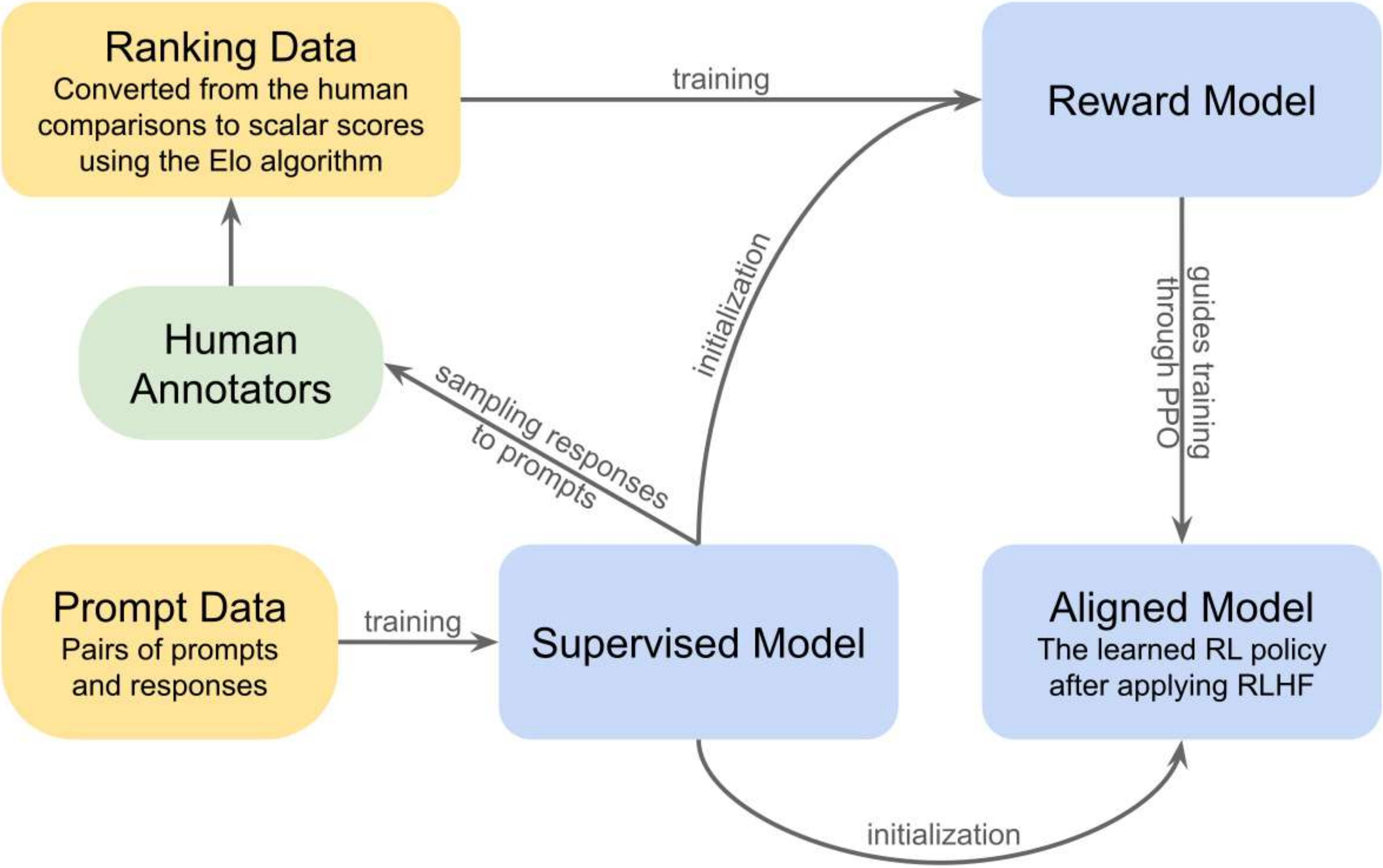
This PR reverts the broken caching logic and restores the original direct product catalog call, which will resolve the latency regression once merged and deployed."


Elastic
Afternoon, Carly
How can I help you today?
Sonnet 4.6
Write
Learn
Code
From Calendar
From Gmail

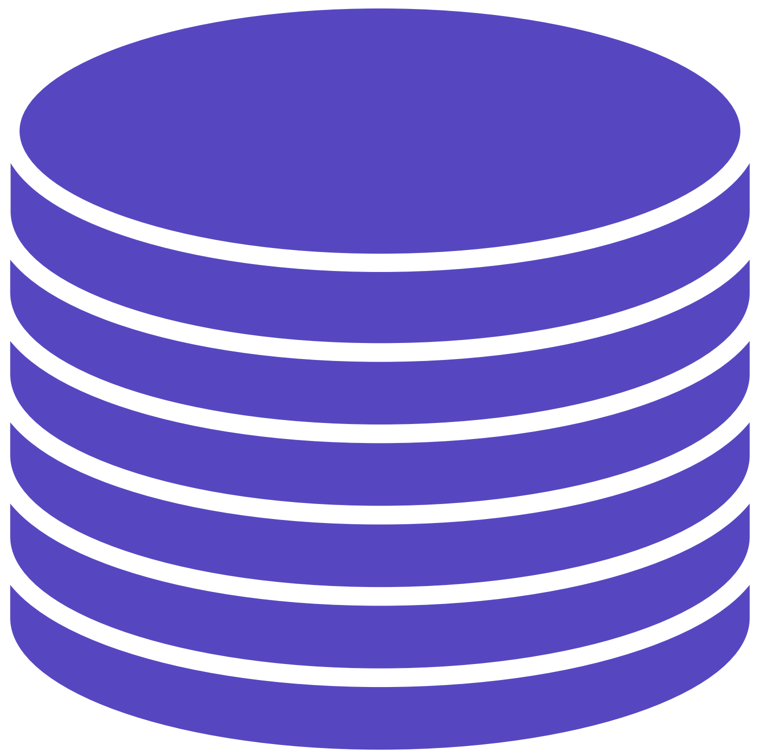
Source: Amy Humphries on Unsplash

THERE'S MORE HUMANS AND LOOPS THAN YOU THINK...

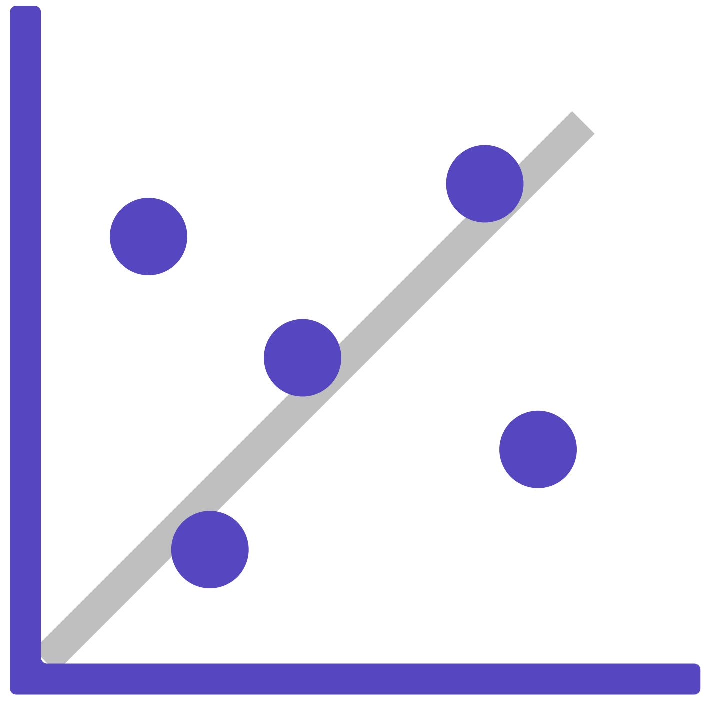
REINFORCEMENT LEARNING WITH HUMAN FEEDBACK



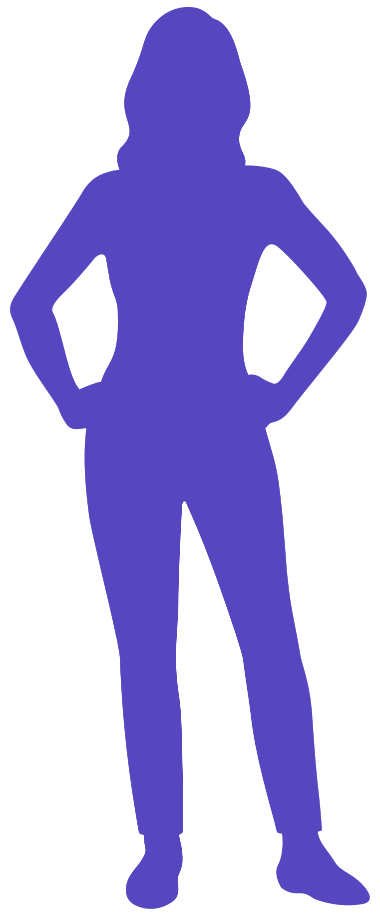
WHY DO LLMS HALLUCINATE?



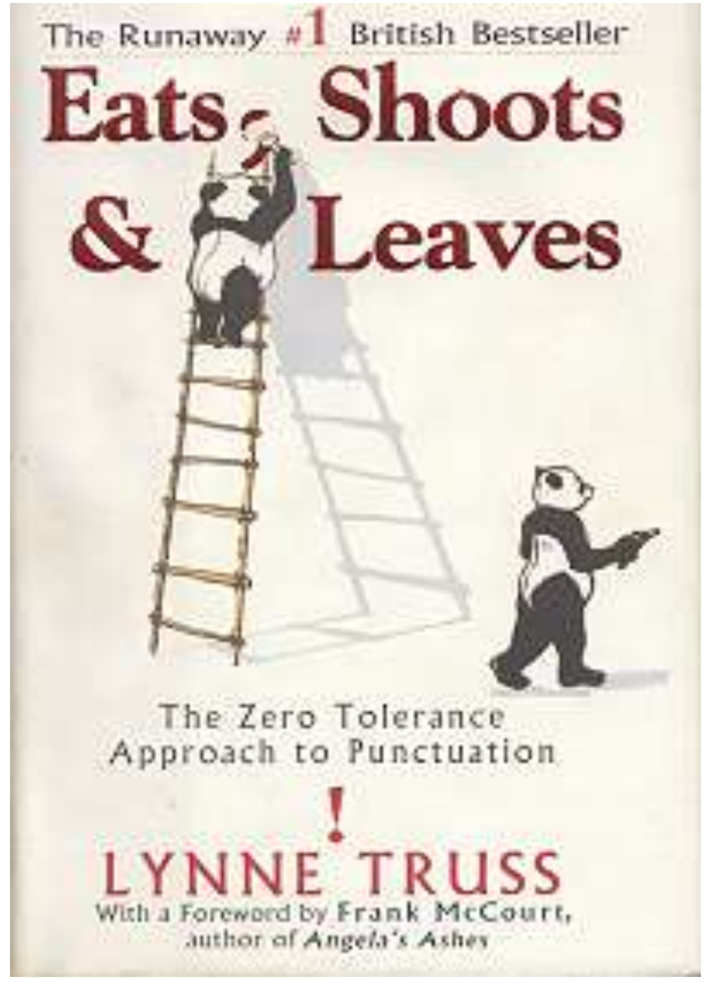
Frozen or Limited Knowledge



Overfitting



Biases



Language Ambiguity

Catastrophic Forgetting

Source: OpenAI | Why Language Models Hallucinate

Why Language Models Hallucinate

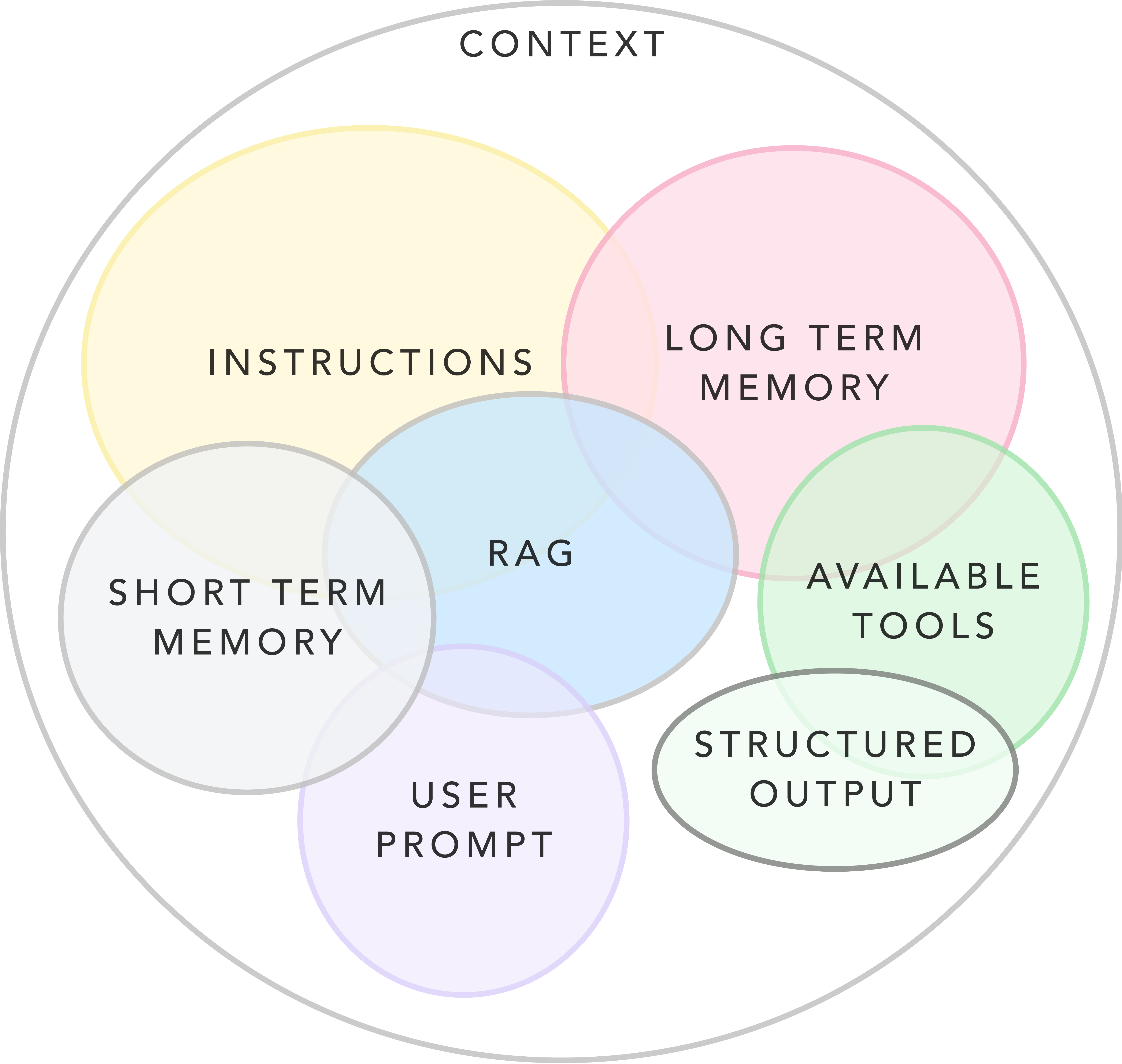
Adam Tauman Kalai* OpenAI
Ofir Nachum OpenAI
Santosh S. Vempala† Georgia Tech
Edwin Zhang OpenAI

September 4, 2025

Abstract

Like students facing hard exam questions, large language models sometimes guess when uncertain, producing plausible yet incorrect statements instead of admitting uncertainty. Such "hallucinations" persist even in state-of-the-art systems and undermine trust. We argue that language models hallucinate because the training and evaluation procedures reward guessing over acknowledging uncertainty and we analyze the statistical causes of hallucinations in the modern training pipeline. Hallucinations need not be mysterious—they originate simply as errors in binary classification. If incorrect statements cannot be distinguished from facts, then hallucinations in pretrained language models will arise through natural statistical pressures. We then argue that hallucinations persist due to the way most evaluations are graded—language models are optimized to be good test-takers, and guessing when uncertain improves test performance. This "epidemic" of penalizing uncertain responses can only be addressed through a socio-technical mitigation: modifying the scoring of existing benchmarks that are misaligned but dominate leaderboards, rather than introducing additional hallucination evaluations. This change may steer the field toward more trustworthy AI systems.

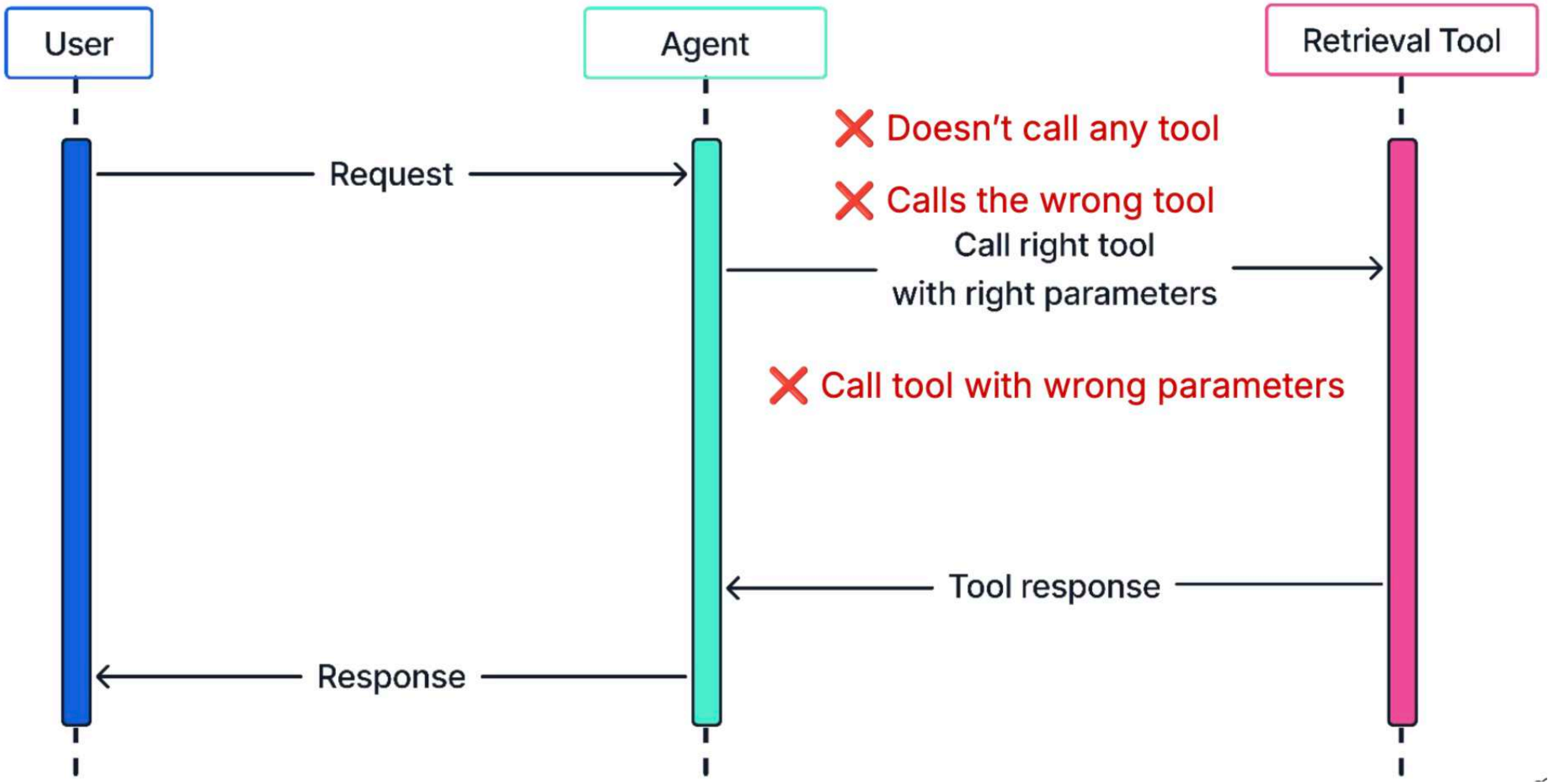
Credit: What is Context Engineering | Data Science Dojo




Source: Hunter Haley on Unsplash
...BUT IN AIOPS HUMANS DETERMINE THE TOOLS
elastic
carlyrichmond.bsky.social


otel-demo
Agent Builder
Agents
observability_custom.agent
AI Agent
AW
Observability
Discover
Dashboards
Workflows
Alerts
Cases
SLOs
Streams
More
Custom Observability Agent
Chat
Save
Agent specialized in logs, metrics, and traces
Settings
Tools 32
This agent has 32 active tools
Tools enable agents to work with your data. For best results, keep the selection under 24 to avoid overwhelming your agent with too many options.
Active tools
32/24
Search
Labels
Show active only
Showing 1-43 of 43 Tools
ID
Labels
platform.core.search
A powerful tool for searching and analyzing data within your Elasticsearch cluster.
platform.core.get_document_by_id
Retrieve the full content (source) of an Elasticsearch document based on its ID and index name.
platform.core.execute_esql
Execute an ES|QL query and return the results in a tabular format.
platform.core.generate_esql
Generate an ES|QL query from a natural language query.
Cancel
Chat
Save

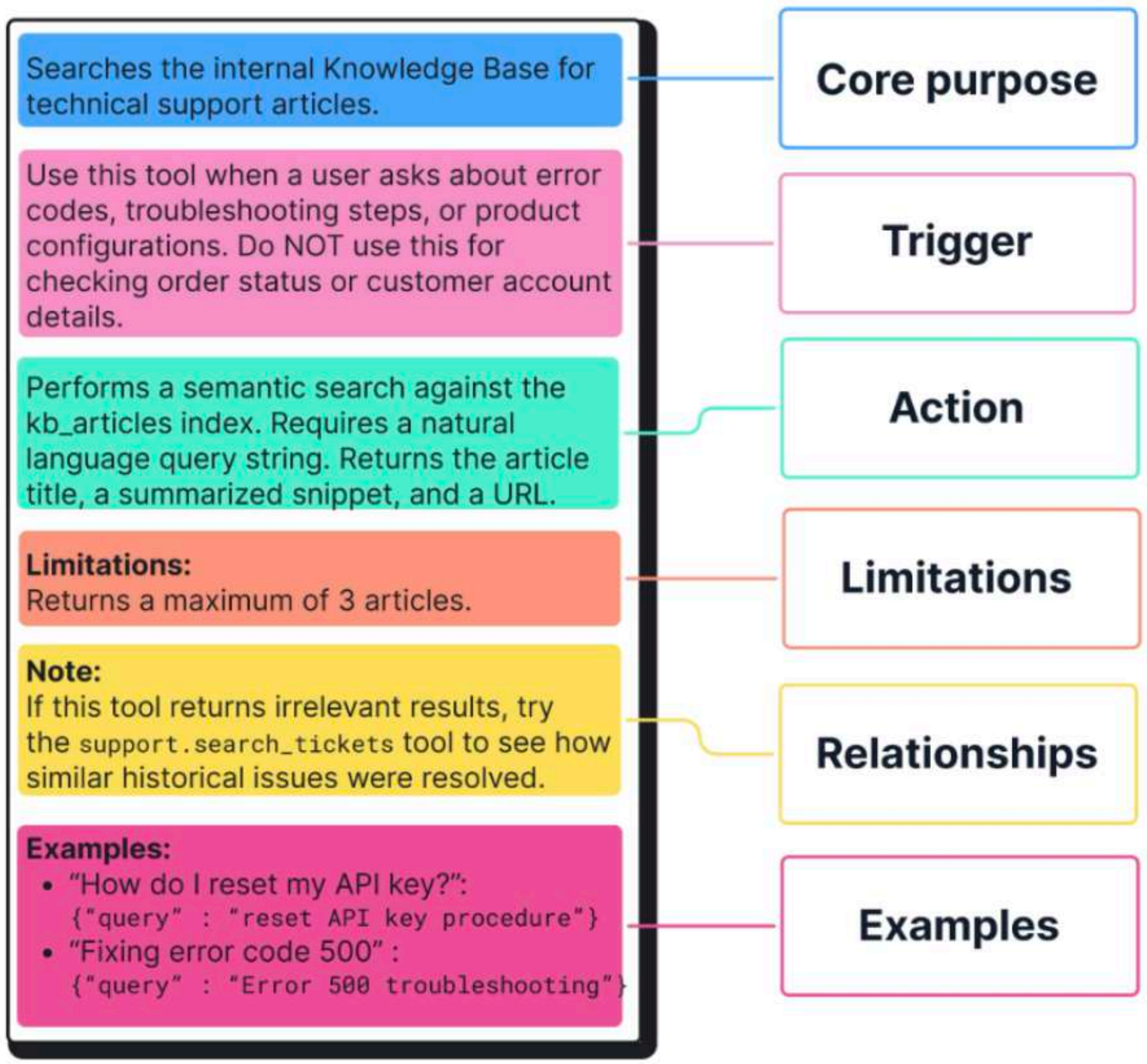
Agentic Tool Challenges



How to make agents call the correct tool

1. Prompt engineer the **tool description**

2. Reinforce in the agent's **system prompt**



EVEN AI GENERATED ALERTS CAN BE TIRING

OTel Pal APP 2:34 PM

"🚨 **Latency Alert: frontend-proxy GET /api/recommendations**

Problem: The `frontend-proxy` service is experiencing elevated latency (318 ms avg, threshold: 200 ms) on the `GET /api/recommendations` transaction.

Root Cause: Commit `a5343cd` in the `recommendation` service introduced a faulty caching function that causes exponential list growth and always executes the expensive path on every request, resulting in unbounded memory consumption and CPU overhead.

Case: https://otel-demo-d90c84.kb.europe-west3.gcp.elastic.cloud/app/observability/cases/6a5f737d-ba66-4f4d-bfb3-59ac3d37d182

Automatic Remediation: A PR has been created to fix this issue:
https://github.com/AlexanderWert/opentelemetry-demo/pull/7

This PR reverts the broken caching logic and restores the original direct product catalog call, which will resolve the latency regression once merged and deployed."

ALERT FATIGUE BEST PRACTICES ARE STILL ALIVE

Source: Various on Unsplash

elastic

carlyrichmond.bsky.social

WHAT WE COVERED

- What is HITL?
- Reinforcement Learning with Feedback
- Context Engineering
- Alert Fatigue management

Source: White.Rainforest ™ ∙ 易雨白林. on Unsplash


T1 H4 A1 N1 K5
Y4 O1 U1